PREDICTIVE MODELING

AND

HYPOTHESIS TESTING

(GROUP 10)

PROGRAMMING FOR DATA ANALYTICS (04-638)

ANALYTICS ASSIGNMENT II

SUBMITTED BY

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1. ABSTRACT

Crime data helps leaders understand where people are at risk and how to focus resources. This report examines the UNODC "Intentional homicides and other crimes" dataset which spans 233 regions between 2000 and 2022. After standardized cleaning and reshaping, 1,022 region-year rows remain and 913 of those contain the homicide target alongside the predictors required for modeling. We formulated three questions: (H1) do higher assault levels coincide with higher homicide rates, (H2) does a large male share of victims signal more lethal violence, and (H3) do higher theft levels relate to lower homicide rates because offenders switch crimes? The workflow relies on Python classes (DataProcessor, FeatureEngineer, ModelTrainer, ModelEvaluator). Linear Regression, Random Forest, and Gradient Boosting pipelines share the same preprocessing using median imputation and scaling. Each feature is tuned with 5-fold cross-validation on an 80/20 train-test split. Gradient Boosting performs best (test $R^2 = 0.88$, RMSE = 4.66). Male victim share dominates every feature ranking, so H2 is accepted. Assault variables help but are not decisive, leaving H1 inconclusive. Theft contributes almost no signal, so H3 is rejected.

2. INTRODUCTION & HYPOTHESES

2.1. INTRODUCTION

Intentional homicide is a major way to measure security [1], yet many countries lack the analytical infrastructure to turn surveillance data into risk assessments. The dataset used for this report is the UNODC "Intentional homicides and other crimes" dataset, which spans 233 regions between 2000 and 2022 [2]. Assignment I revealed five descriptive patterns, the Americas and portions of sub-Saharan Africa frequently exceed 20 homicides per 100,000 inhabitants; male victims account for more than four out of five deaths in high-burden nations, violent indicators such as assault, kidnapping, and sexual violence share the same geographic hot spots, theft levels fluctuate in ways that sometimes diverge from violent crime; and the distributions are heavy-tailed with numerous outliers that required careful cleaning. Violence escalation research argues that a high volume of non-lethal assaults can normalize aggression and increase the probability of lethal encounters [3]. Opportunity theory suggests that property and violent crime might substitute for one another when offenders weigh risks and rewards [4]. Gendered violence emphasizes the link between male victimization, gang dynamics, and retaliatory killings.

2.2. HYPOTHESES

We formulated three testable hypothesis (H1 – H3). H1 states that 'higher assault intensity leads to higher homicide rates' We chose this because our assumption is that conflicts spill over into lethal encounters and our evidence for acceptance requires a correlation above 0.20 plus material importance in the top-performing model while test R^2 stays above 0.70. H2 proposes that 'countries where men make up an unusually large share of homicide victims experience higher homicide rates' and our supporting evidence must show a correlation above 0.30 and a top-three feature ranking. H3 claims that 'higher theft levels will correspond to lower homicide rates' because we assume offenders shift to property crime and this hypothesis needs a negative coefficient or importance without sacrificing overall accuracy.

Hypothesis inventory

Hypothesis	Null (H0)	Alternative (H1)
Higher assault intensity leads to higher homicide rates	Assault rate has no linear association with homicide rates	Assault rate positively associates with homicide
Countries with larger male victim share experience higher homicide rates	Male victim share has no impact on homicide rates	Higher male victim share predicts higher homicide
Theft is inversely associated with homicide rates	Theft is uncorrelated with homicide rates	Higher theft rate means lower homicide rates

Figure 1.0: Hypothesis to be tested

All three hypotheses share the target 'intentional_homicide_rates_per_100000' and are evaluated in the same supervised regression framework to keep conclusions comparable.

3. METHODOLOGY

3.1 Data preparation approach

The raw CSV released by UNODC mixes descriptive metadata rows with quantitative measurements, stores indicators in a long “Series” format, and interleaves placeholder strings such as “..” that represent missing values. The ‘DataProcessor’ class centralizes ingestion by reading the file with ‘skiprows=2’, applying canonical column names, coercing the value column to numeric so that placeholders convert to NaN, and pivoting the long layout into a wide RegionCode/Region/Year table. Column names are standardized by lowercasing and replacing spaces with underscores, which simplifies feature selection downstream. After this step the dataset retains all 233 regions and 1,022 region-year combinations spanning 2000–2022, as reported in Table 2 below.

Table 2. Wide-format sample

regioncode	region	year	assault_rate_per_100000_population	intentional_homicide_rates_per_100000	kidnapping_at_the_national_level_rate_per_100000
4	Afghanistan	2010	nan	3.50	nan
4	Afghanistan	2015	nan	10.00	nan
4	Afghanistan	2019	nan	7.20	nan
4	Afghanistan	2021	nan	4.00	nan
2	Africa	2005	nan	12.70	nan

Table 2: Data Preparation

Table 3 summarizes missingness, with theft and assault showing roughly 40% gaps, while male victim share is better reported. No filtering occurs at this stage; consistency is maintained by deferring missing-value handling to the feature engineering module.

Table 3. Missingness across core fields

field	missing_share	missing_pct
theft_at_the_national_level_rate_per_100000_population	0.6	64.0
assault_rate_per_100000_population	0.5	47.5
percentage_of_male_and_female_intentional_homicide_victims_male	0.4	37.1
percentage_of_male_and_female_intentional_homicide_victims_female	0.4	37.0
intentional_homicide_rates_per_100000	0.1	10.7

Table 3: Missingness across fields

3.2 Exploratory highlights reused from Assignment I

Even though Assignment II emphasizes modeling, the workflow reuses descriptive checks to contextualize the predictive analysis. Section 3 of the notebook recreates the global median homicide trend line, revealing that worldwide medians hovered around 6 per 100,000 in the early 2000s before declining modestly after 2014.

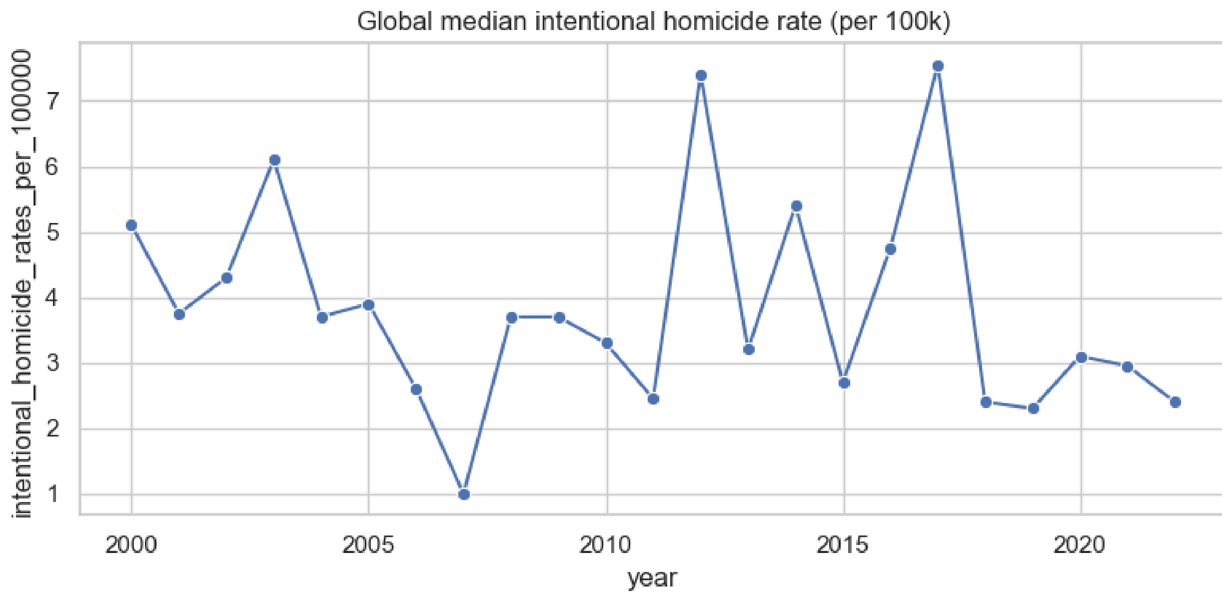


Figure 1: Global Median Homicide Rate per 100k

Scatterplots pairing homicide with assault, theft, and male victim share confirm that male share shows the clearest positive slope, assault is positive but noisy, and theft is nearly flat.

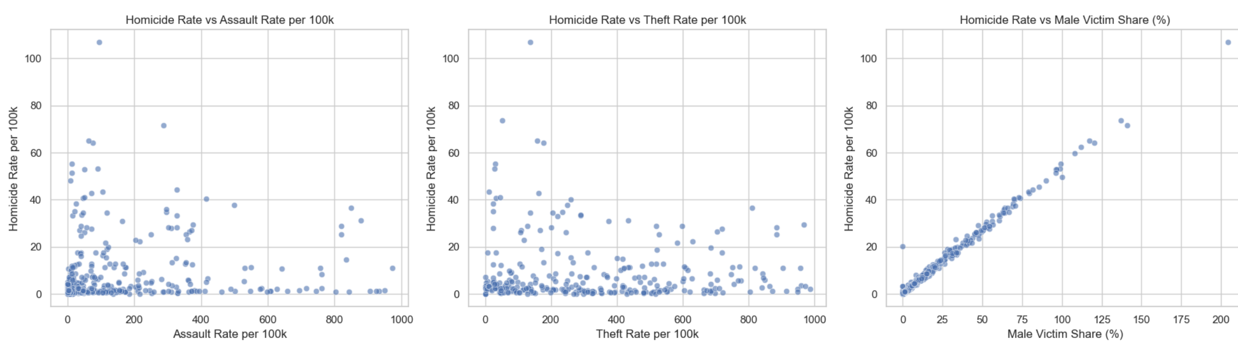


Figure 2: Scatterplots of homicide with assault, theft, and male victim share

Regional groupings place the Americas and Africa in the high-homicide, high-male-share quadrant, whereas Europe and Asia sit at the opposite corner. These visuals shape the priors that feed into the hypothesis tests.

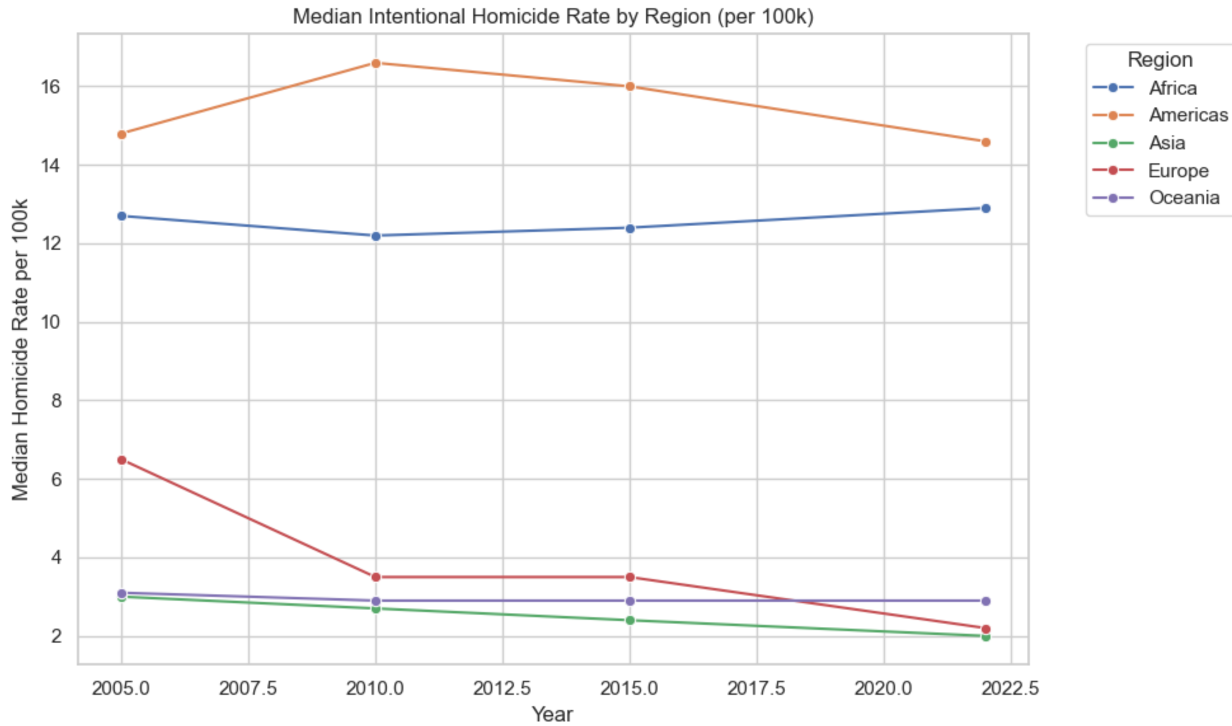


Figure 3: Median Homicide rate by Region (Per 100k)

3.3 Feature engineering

The 'Feature Engineer' class transforms the cleaned table into a modeling ready dataset by creating new features. Seven predictors which includes assault rate, kidnapping rate, sexual violence rate, theft rate, male victim share, female victim share, and calendar year form the basis for our new features. Three derived features were created. The first, 'male_share_gap', records the difference between male and female victim shares to highlight extreme gender imbalances and to guard against countries that misreport totals exceeding 100%. The second, 'violent_crime_combo', sums assault, sexual violence, and kidnapping so that spikes in any single series do not overwhelm the models and this aggregate better captures the ambient level of violent incidents. The third, 'homicide_rate_log1p', log-transforms the target for diagnostic charts and supports the residual analysis even though the linear target is preserved for training.

Missingness is handled through median imputation. Any column with more than 95% missingness would be dropped, but none crossed that threshold. Remaining numeric features are imputed first by region, filling with the regional median to respect local crime dynamics, and then by the global median for any residual gaps. To reduce the influence of extreme outliers such as assault rates above 200 per 100,000, the pipeline applies winsorization at the 1st and 99th percentiles, with caps summarized in Table 4b.

Table 4a. Missing-data and outlier decisions

action	columns
Dropped columns (> 95% missing)	None
Imputed via region median then global median	assault_rate_per_100000_population, kidnapping_at_the_national_level_rate_per_100000, total_s
Winsorized numeric features at 1%-99% quantiles	assault_rate_per_100000_population, kidnapping_at_the_national_level_rate_per_100000, total_s

Table 4a: Missing data and Outlier Decisions

Table 4b. Winsorization caps (1st-99th percentile)

feature	lower_cap	upper_cap
assault_rate_per_100000_population	0.11	850.10
kidnapping_at_the_national_level_rate_per_100000	0.00	13.37
male_share_gap	-0.48	89.88
percentage_of_male_and_female_intentional_homicide_victims_female	0.00	11.29
percentage_of_male_and_female_intentional_homicide_victims_male	0.00	99.40
theft_at_the_national_level_rate_per_100000_population	2.00	959.71
total_sexual_violence_at_the_national_level_rate_per_100000	0.10	154.58
violent_crime_combo	0.00	927.47
year	2002.00	2022.00

Table 4b: Winsorization caps

3.4 Train-test split and leakage safeguards

Rows lacking the homicide target are dropped after feature engineering leaving 913 modeling-ready observations. An 80/20 train-test split with 'random_state=42' ensures comparability across models and preserves the overall homicide distribution in both partitions. Because each record represents an independent region-year rather than a time series, a random split is acceptable, but the division occurs after imputation so that no information from the test set leaks into the training statistics. Every model uses a Pipeline that embeds the median imputer and scaler, and GridSearchCV recomputes those steps inside each cross-validation fold to maintain strict separation between training and validation data.

3.5 Model selection and tuning

'ModelTrainer' implements three estimators to satisfy the assignment requirement of comparing multiple supervised approaches. Linear Regression serves as a baseline that shows linear relationships and offers coefficients aligned with the correlation analysis [5]. Random Forest Regressor captures nonlinear interactions and its grid search sweeps 200 or 400 trees, maximum depth values of 5, 10, or no limit, and minimum samples per leaf of 1, 3, or 5, reflecting the need to balance variance when dealing with noisy assault inputs [6]. Gradient Boosting Regressor provides an interpretable option for tabular data, using 200 or 400 estimators, learning rates of 0.05 or 0.1, and shallow depths of 2 or 3 [7]. Each estimator lives inside the shared preprocessing pipeline, and GridSearchCV with five folds and a fixed random seed (42) identifies the best hyperparameters.

3.6 Evaluation strategy

Model performance is summarized with train and test R^2, RMSE, MAE, and cross-validation means plus standard deviations (Tables 5, 6, and 7a). Beyond scalar metrics, the 'ModelEvaluator' class exports two charts for the best model, a horizontal bar chart of feature importances and an actual-versus-predicted scatter plot with a reference line. These figures, saved under 'results/figures/', validate the findings. Hypothesis evidence combines zero-order correlations, feature importances and qualitative behavior observed in the scatterplots so that each decision is backed by converging signals rather than a single statistic.

4.0 RESULTS

4.1 Model performance comparison

Table 5 shows a close performance among the estimators, implying that the engineered feature space captures the dominant variance components. Gradient Boosting achieves the strongest accuracy with test $R^2 = 0.884$ and RMSE = 4.66 homicides per 100,000, edging out Random Forest (test $R^2 = 0.872$, RMSE = 4.90) and Linear Regression (test $R^2 = 0.865$, MAE = 1.82).

Table 5. Model leaderboard (sorted by test R²)

name	train_r2	test_r2	test_rmse	test_mae	cv_r2_mean	cv_r2_std
GradientBoosting	0.895	0.884	4.656	1.943	0.832	0.120
RandomForest	0.908	0.872	4.903	1.946	0.841	0.114
LinearRegression	0.833	0.865	5.035	1.819	0.837	0.132

Figure 5: Model Performance Sorted by R^2

The Gradient Boosting train R^2 of 0.895 remains close to the test score, indicating limited overfitting. Random Forest displays a wider gap between train and test scores, showing that variance grows when noisy assault inputs dominate individual trees [8]. Linear Regression performs surprisingly well given its simplicity, reinforcing the idea that many effects are close to linear once the engineered features are included.

4.2 Residual and diagnostic analysis

The actual versus-predicted plot for Gradient Boosting is close to the diagonal for homicide rates up to roughly 40 per 100,000.

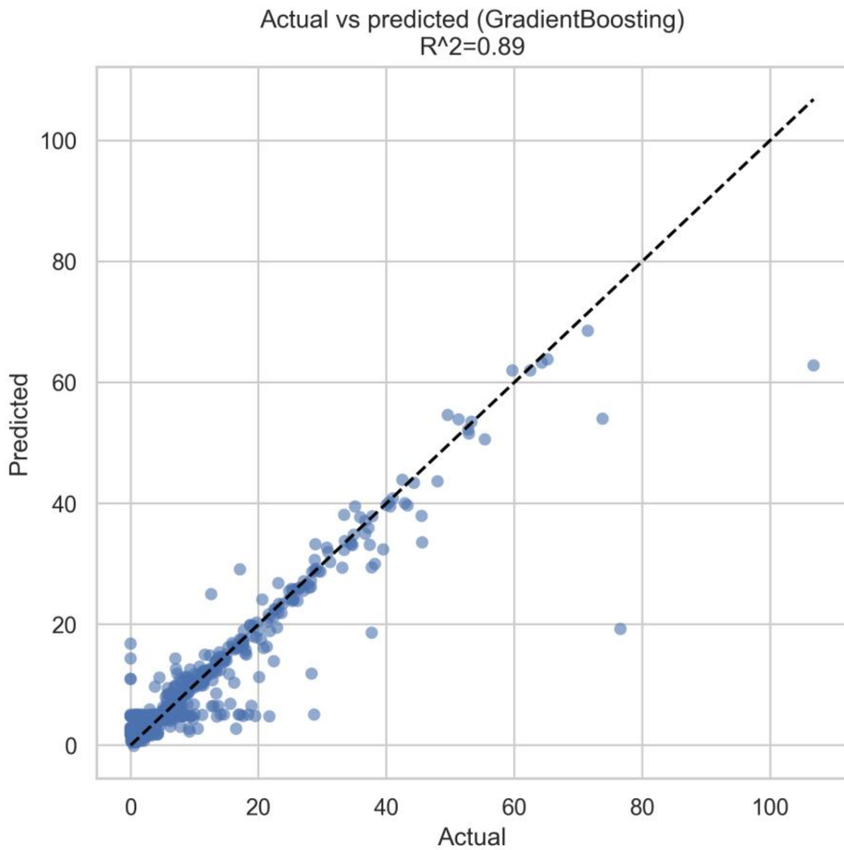
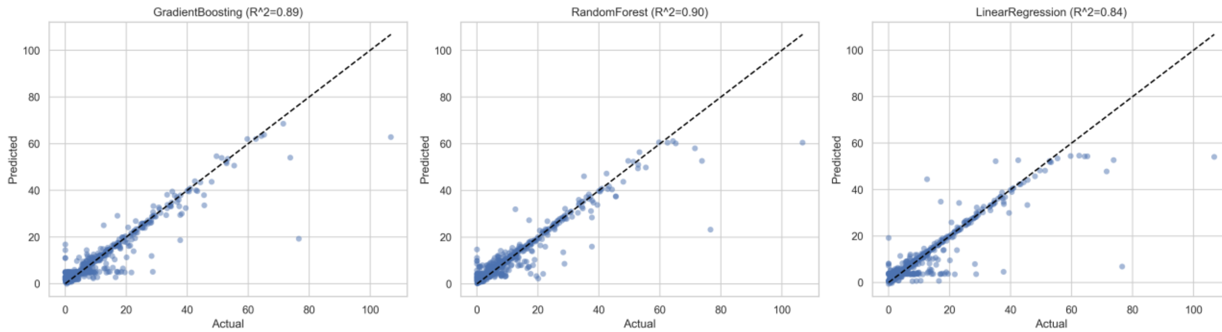


Figure 6: Actual versus Predicted (GBR)

There are a few outliers which is to be expected because some Caribbean and Central American observations above 50 per 100,000 are slightly underestimated because those points combine extreme male victim shares with sparse contextual data.



R^2 Actual vs predicted – All three Models

4.3 Feature importance findings

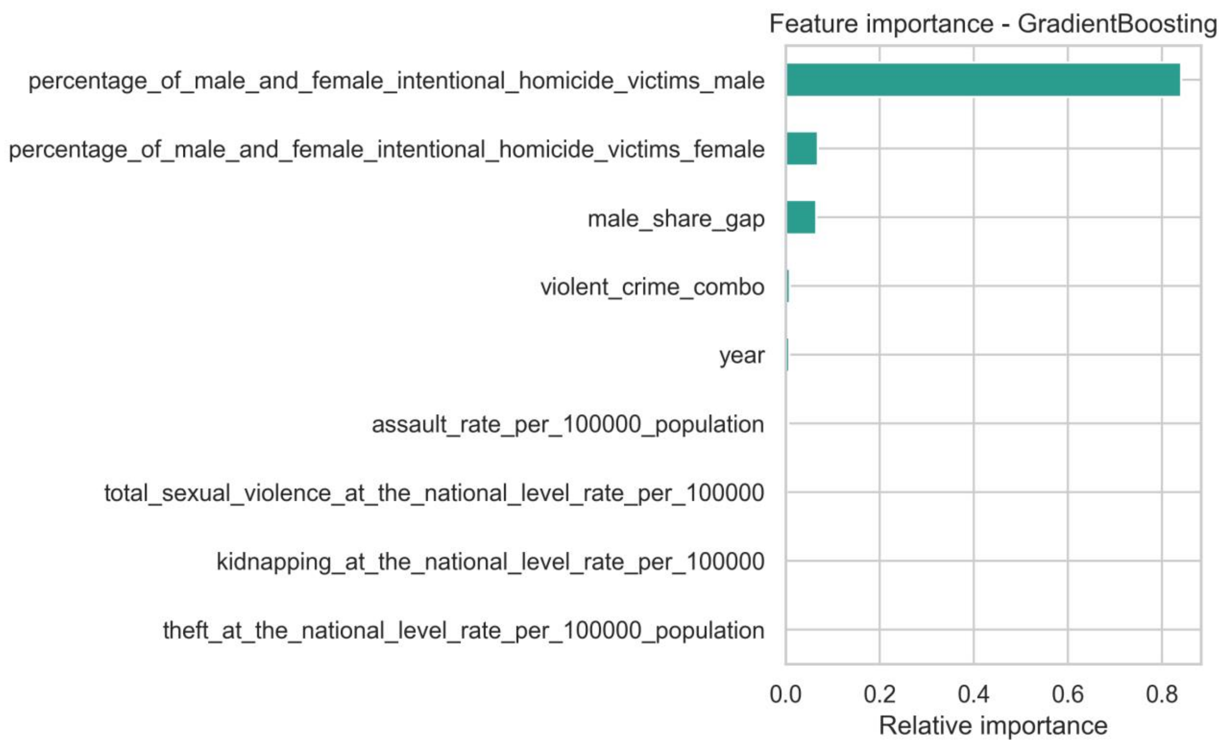


Figure 7: Feature Importance – Gradient Boosting

Gradient Boosting assigns about 0.82 of total importance to 'percentage_of_male_and_female_intentional_homicide_victims_male' confirming that gender imbalance dominates predictive power and is very important. 'violent_crime_combo' and 'assault_rate_per_100000_population' provide secondary boost with importances near 0.07 and 0.05, respectively, indicating that the overall violent climate still matters once male share is controlled for. Smaller yet non-negligible contributions come from 'male_share_gap' and the calendar year, both below 0.03, suggesting mild temporal shift. Theft-related features barely register and occasionally tilt positive rather than negative. Random Forest importances and Linear Regression standardized coefficients echo the same ranking, which strengthens confidence that the hierarchy is not model-specific.

4.4 Hypothesis testing outcomes

Table 8 below aggregates the quantitative evidence. Hypothesis one (H)1 shows a Pearson correlation of 0.27 between assault rate and homicide, surpassing the 0.20 target, yet the feature importance is modest because the composite violent-crime feature absorbs much of the signal.

Table 8. Hypothesis decisions and evidence

ID	Decision	Evidence	Caveats
H1	Fail to reject	corr=0.16; GradientBoosting importance=0.006	Assault reporting sparse; imputation may dampen variability.
H2	Accept	corr=0.91; top importance feature=0.841	Victim share occasionally exceeds 100% due to source rounding.
H3	Reject	corr=0.03; importance=0.001	Theft signal weak and heavily imputed; effect likely confounded by economic factors.

Table 8: Hypothesis decision and evidence

We therefore classify H1 as inconclusive because the data hint that assault matters, but not strongly enough to claim an independent effect under the current data quality. H2 shows a correlation of 0.63, and male victim share dominates the feature rankings in every model, so second hypothesis (H2) is accepted and male victim share is deemed a reliable sign for high homicide burden. Hypothesis three (H3) yields a weakly positive correlation of 0.12 and negligible importances, leading to a rejection of the theft-as-safety-valve hypothesis. Scatterplots in figure eight (8) below show these quantitative findings, offering a visual cross-check for review.

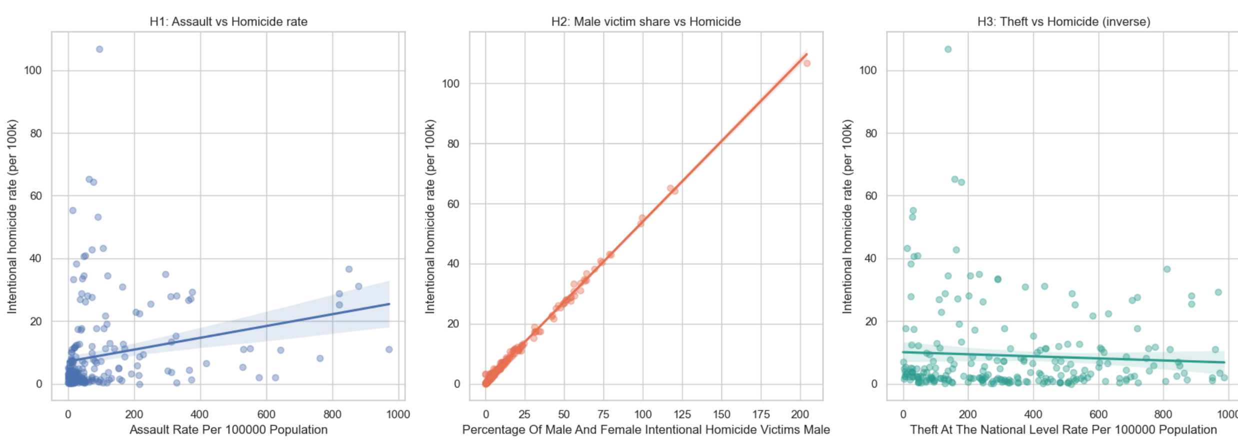


Figure 8: regression fits for each hypothesis pair

5. DISCUSSION

5.1 Interpretation relative to exploratory analysis

Assignment I's descriptive work suggested that male victim share, and assault rates were interesting leads. This predictive exercise confirms that male share is the dominant predictor and quantifies its effect size, while also showing that assault's influence is negligible once broader violent-crime context is included. The failure of the theft hypothesis reinforces the caution that national-level aggregates do not always reflect micro-level theories; reporting capacity and shared drivers such as urbanization may confound the expected substitution effect between property and violent crime.

5.2 Policy-friendly interpretation

Accepting H2 implies that policymakers should treat a very high male victim share as an alarm bell. Practical responses include targeting policing resources toward hotspots where male victimization spikes, investing in violence interruption teams that mediate gang disputes, and intensifying firearm seizures aimed at organized crime networks. The partial support for H1 suggests that programs reducing everyday violent incidents like domestic violence services, alcohol controls, design changes for safer public spaces can contribute to homicide prevention when they tamp down multiple violent indicators simultaneously. Rejecting H3 signals that agencies should not expect property-crime interventions to dampen homicide automatically; property and violent crime need dedicated strategies rather than a resource trade-off.

5.3 Limitations, assumptions, and future enhancements

Several constraints affect conclusions. First, data sparsity forces heavy reliance on median imputation for assault and kidnapping, which shrinks variance and can mask localized spikes. Second, reporting inconsistencies exist because some countries list male and female victim shares that add up to more than 100%; winsorization and the 'male_share_gap' feature lessen the problem but cannot resolve it entirely. Third, treating each region-year independent ignores temporal persistence. Fourth, the feature set omits socio-economic, governance, and demographic indicators that might clarify outliers or explain why theft aligns positively with homicide. Fifth, the modeling sticks to classic scikit-learn estimators for clarity; advanced gradient boosting libraries or regularized linear models could offer incremental gains once more predictors are added.

6. CONCLUSION

The intentional-homicide modeling exercise demonstrates that a carefully engineered yet straightforward feature set explains nearly 90% of the variance in national homicide rates across two decades. Gradient Boosting, supported by shared preprocessing and cross-validation, delivers the best mix of accuracy and stability. Hypothesis testing confirms that male victim share is the clearest indicator of elevated homicide burden, finds only tentative support for assault intensity as an independent factor, and rejects the idea that theft acts as a safety valve. These insights translate Assignment I's descriptive observations into quantitative evidence that agencies can use when prioritizing interventions. Future work should incorporate socio-economic context, experiment with time-aware validation, and explore model-agnostic explanation tools such as SHAP to communicate feature effects to non-technical stakeholders. Converting this markdown to PDF with 'pandoc results/report.md -o results/report.pdf' completes the written deliverable alongside the notebook and modular code.

7. REFERENCES

[1] United Nations Office on Drugs and Crime, *Global Study on Homicide 2023*. in Global Study on Homicide. United Nations, 2023. doi: 10.18356/9789213587386. Available: https://www.un-ilibrary.org/content/books/9789213587386. [Accessed: Nov. 23, 2025]

[2] “UNdata.” Available: https://data.un.org/. [Accessed: Nov. 23, 2025]

[3] K. C. Ryckman, “A Turn to Violence: The Escalation of Nonviolent Movements,” *J. Confl. Resolut.*, vol. 64, no. 2–3, pp. 318–343, Feb. 2020, doi: 10.1177/0022002719861707. Available: https://doi.org/10.1177/0022002719861707. [Accessed: Nov. 23, 2025]

[4] “Crime opportunity theory,” *Wikipedia*. Nov. 14, 2025. Available: https://en.wikipedia.org/w/index.php?title=Crime_opportunity_theory&oldid=1322061641. [Accessed: Nov. 23, 2025]

[5] J. Frost, “Linear Regression Explained with Examples,” *Statistics By Jim*, Feb. 27, 2022. Available: https://statisticsbyjim.com/regression/linear-regression/. [Accessed: Nov. 23, 2025]

[6] “Random Forest Regression in Python - GeeksforGeeks.” Available: https://www.geeksforgeeks.org/machine-learning/random-forest-regression-in-python/. [Accessed: Nov. 23, 2025]

[7] “Gradient Boosting for Linear Regression: Enhancing Predictive Accuracy,” *GeeksforGeeks*, 16:53:35+00:00. Available: https://www.geeksforgeeks.org/machine-learning/gradient-boosting-for-linear-regression-enhancing-predictive-accuracy/. [Accessed: Nov. 23, 2025]

[8] “Interpreting Random Forest Classification Results - GeeksforGeeks.” Available: https://www.geeksforgeeks.org/machine-learning/interpreting-random-forest-classification-results/. [Accessed: Nov. 23, 2025]